

Quantifying Building Blocks of Life in Planetary Analog Materials: Implications for Prebiotic Chemistry and Biosignature Identification

Xiao'ou Luo¹, Chao He^{1,*}, Zhengbo Yang¹, Yingjian Wang¹, Ziyao Fang¹, Yu Liu¹, Sai Wang¹, Haixin Li¹

1. National Key Laboratory of Deep Space Exploration / School of Earth and Space Sciences, University of Science and Technology of China, Hefei 230026, China

*Corresponding authors: Chao He (chaohe23@ustc.edu.cn)

Abstract

Building blocks of life such as amino acids, nucleobases, and fatty acids are central to prebiotic chemistry and represent key targets in the search for planetary biosignatures. In planetary materials, biomolecules typically occur at trace levels within complex matrices, posing substantial analytical challenges, particularly for quantitative characterization. Here we develop a gas chromatography–tandem mass spectrometry (GC/MS/MS) method that enables robust qualitative and quantitative analysis of 56 prebiotically relevant molecules. The method is applied to a Titan aerosol analog and, for the first time, to a Martian gypsum analog from the Qaidam Basin, revealing diverse inventories of amino acids, nucleobases, and fatty acids in both samples. In the Titan aerosol analog, the first detection of phenylalanine and an extensive inventory of fatty acids, together with elevated nucleobase abundances, offers new insights into atmospheric photochemical synthesis of prebiotic molecules. In the Martian analog sample, amino acids are detectable and exhibit pronounced biotic–abiotic contrasts in abundance patterns relative to those observed in the Titan aerosol analog, whereas fatty acids show more overlapping abiotic and biotic signatures, highlighting the potential of amino acids as robust biosignatures. These results provide quantitative constraints on prebiotic chemical evolution and underscore the utility of GC/MS/MS for biosignature identification in planetary exploration.

Keywords: GC/MS/MS; Prebiotic Chemistry; Biosignatures; Titan Aerosol Analogs; Martian Analog Samples

1. Introduction

The investigation of life's origins and the search for extraterrestrial life are fundamental objectives of planetary exploration. The origin of life on Earth is generally thought to have started from exogenously and endogenously produced prebiotic organic building blocks, including amino acids, nucleobases, and fatty acids. These molecules can be synthesized on Earth through various abiotic pathways such as atmospheric chemistry (Miller 1953; Miller & Urey 1959; Hörst et al. 2012; Ferus et al. 2017; Sebree et al. 2018; Pearce et al. 2024a) and hydrothermal reactions (McCollom et al. 1999; LaRowe & Regnier 2008; Herschy et al. 2014; Zhang et al. 2017; Marlin et al. 2025), or they may be delivered by meteorites (Chyba & Sagan 1992; Burton et al. 2012; Pearce & Pudritz 2015; Naraoka et al. 2023; Glavin et al. 2025). Hypotheses for the origin of life suggest that under favorable conditions, these molecules can polymerize into biological macromolecules (e.g., proteins, RNA, DNA), eventually giving rise to primitive forms of life. However, finding evidence for such prebiotic processes on early Earth is challenging due to the lack of direct geological records. An alternative approach is to study similar ongoing processes on other planetary bodies. For example, on Saturn's moon Titan, photochemistry in the upper atmosphere produces complex organic aerosols. Investigating the formation of life's building blocks in such environment can deepen our understanding of prebiotic chemical evolution. Additionally, the identification of these biomolecules in planetary materials can also aid the search for extraterrestrial life. Compared to more complex and unstable biological macromolecules, these smaller compounds are technically easier to detect and have a greater potential for preservation over long periods. Consequently, they may serve as biosignatures, providing evidence of extinct or present life on potentially habitable planets such as Mars.

Therefore, detecting these building blocks of life, regardless of their abiotic or biotic origin, provides valuable insights into the evolution of life and has become a key focus in planetary exploration missions. Their analysis is typically carried out using gas or liquid chromatography coupled with mass spectrometry (e.g., quadrupole, Orbitrap, or time-of-flight mass spectrometry). Among these techniques, gas chromatography–mass spectrometry (GC/MS) is widely employed owing to its high separation efficiency and access to extensive reference mass spectral libraries. Several planetary missions have carried GC/MS instruments for in-situ analysis, aiming to detect biomolecules on Mars, Comet 67P, and Titan. Although biomolecules have not been confirmed on these bodies, a variety of other organic molecules have been detected (Niemann et al. 2005; Freissinet et al. 2015; Altwegg et al. 2017; Freissinet et al. 2025). In laboratories, where advanced instruments and optimized sample preparation procedures are available, numerous biomolecules have been successfully identified in planetary materials or their analogs using GC/MS. Meteorites represent the most extensively studied planetary materials, in which amino acids (Kvenvolden et al. 1970; Engel & Nagy 1982), nucleobases (Martins et al. 2008), carboxylic acids (Yuen et al. 1984; Huang et al. 2005), and carbohydrates (Cooper et al. 2001; Furukawa et al. 2019) have been identified and quantified. Some of these molecules, including five nucleobases (cytosine, adenine, thymine, guanine, and uracil) and two amino acids (alanine and glycine), have also been detected in planetary analog materials such as laboratory-produced Titan's organic aerosol analogs (Hörst et al. 2012). However, the sensitivity of GC/MS is often limited by background noise. The gas chromatography–tandem mass spectrometry (GC/MS/MS) technique, employing a triple-quadrupole configuration, can improve detection limits by up to two orders of magnitude when operated in selected reaction monitoring (SRM) mode. In this mode, selected parent ions undergo controlled fragmentation to generate diagnostic daughter ions, which are

selectively monitored to suppress background interference. Therefore, this technique provides more structural information than traditional GC/MS and enables more reliable identification of trace species in complex samples.

GC/MS/MS has been widely applied to the analysis of biomolecules in both planetary materials—such as meteorites (Timoumi et al. 2025) and returned asteroid samples (Furukawa et al. 2025; Glavin et al. 2025; Mojarro et al. 2025)—and planetary analog materials including photochemical organic aerosol analogs (Sebree et al. 2018; Pearce et al. 2024a). These studies demonstrate the strong capability of GC/MS/MS for sensitive, reliable detection and quantitative measurement of diverse prebiotic compounds (amino acids, nucleobases, sugars, and other prebiotic species) in complex matrices. Such quantitative information is crucial for understanding prebiotic chemistry and assessing potential biosignatures. First, quantifying the concentrations of these biomolecules can help evaluate the contributions of specific prebiotic sources and determine whether polymerization reactions are feasible under plausible prebiotic conditions. Second, quantitative constraints can also provide important insights into the potential origin of these compounds, as the patterns of their relative abundances can indicate whether they are of abiotic or biotic origin. For example, abiotic processes generally favor the formation of simpler amino acids, whereas more complex amino acids are typically enriched in biotic systems (Dorn et al. 2011). Despite these strengths, quantitative analyses of biomolecular abundances using GC/MS/MS have remained relatively limited in studies of planetary materials and their analogs. Moreover, fatty acids, a key class of prebiotic molecules linked to protocell formation, have not been systematically investigated in such analyses.

To address these gaps, we developed a GC/MS/MS method capable of quantitatively analyzing 56 biomolecules—including amino acids, nucleobases, fatty acids and other prebiotic precursors—using a single, unified protocol. We applied this method to two planetary analog materials, including a Titan aerosol analog sample and a Martian analog sample from the Qaidam Basin, to evaluate its utility for interpreting molecular inventories relevant to prebiotic chemistry and potential biosignatures in planetary environments.

2. Methods

2.1. Sample Preparation

Titan aerosol analogs were prepared using an electron cyclotron resonance (ECR) device, as described in our previous work (Liu et al. 2023; Yang et al. 2025). In this experiment, a gas mixture of N₂, CH₄, and CO (94.8%/5%/0.2%, each with a purity of 99.99%) was introduced into a vacuum chamber at a total gas flow rate of 50 standard cubic centimeters per minute (sccm), with the chamber pressure settled at 15 Pa. The CO fraction was intentionally elevated relative to Titan's atmosphere (~10 ppm) to promote the formation of O-bearing biomolecules and facilitate their detection. The sample serves as a representative case rather than to reproduce Titan-specific chemistry. The gas mixture was exposed to an ECR plasma in the chamber to simulate the energy sources driving photochemical reactions in Titan's upper atmosphere. After 3 hours of reaction, we collected ~1 g of solid-phase products and dissolved 20 mg of them in 2 mL of methanol (HPLC grade). The solution was vortexed for 5 minutes, and then centrifuged at 11000 rpm for 10 minutes to obtain the supernatant for subsequent processing.

Martian analog samples were collected from the Qaidam Basin located in the north of Tibetan Plateau in northwest China, one of the world's largest and driest playas with an average elevation of ~2800 m. Since the uplift of the Tibetan Plateau, the area has undergone progressive aridification and irradiation, conditions that closely resemble the

Martian surface environment (Anglés & Li 2017; Xiao et al. 2017). Gypsum crystals collected from the Xiaoliangshan Dome in the Dalangtan Area (38°28.85', 91°21.60') were used in this study (Li et al. 2024). We first selected a relatively large crystal without visible cracks and cleaned the surface with methanol to minimize surface contamination. Then we crushed it to smaller pieces, collected several interior fragments, and ground them into fine powders. For extraction, 200 mg of the powder was mixed with 500 μ L of methanol, ultrasonicated for 15 minutes at room temperature, and then centrifuged at 11000 rpm for 10 minutes to collect the supernatant. The procedure was repeated twice, and the resulting supernatants were combined. All equipment used in the whole procedure was pre-washed with methanol and baked in an oven at 80 °C overnight.

The sample solutions were derivatized before GC/MS/MS analysis to form silylated derivatives with lower boiling points and enhanced thermal stability, which allows them to volatilize in the GC inlet without decomposition. The derivatization procedure is modified from that used in previous studies (Sebree et al. 2018; Pearce et al. 2024a). 200 μ L of Titan aerosol analog solution and 1.2 mL of the gypsum extract (volumes adjusted based on biomolecule concentrations estimated from preliminary experiments) were added to GC vials and dried at 50 °C under nitrogen (99.999%) flow for 30 minutes. 200 μ L of dichloromethane (DCM, $\geq$ 99.8%, purchased from Sigma-Aldrich) was added to each vial and then dried again at 50 °C under N₂ flow to remove any residual water. For the silylation step, 30 μ L of dimethylformamide (DMF, $\geq$ 99.8%, purchased from Sigma-Aldrich) and 30 μ L of N-tert-butyldimethylsilyl-N-methyltrifluoroacetamide (MTBSTFA, $\geq$ 97%, purchased from Sigma-Aldrich) were added. The vials were sealed after purged by N₂ flow for 30 seconds to avoid evaporation of reagents, followed by 30 minutes of reaction at 80 °C. Finally, 300 μ L of DCM was added to dissolve derivatized products. Two parallel samples were carried through this procedure, along with two procedural blanks containing pure methanol for contamination check.

2.2. GC/MS/MS Analysis

To develop a targeted SRM method for analyzing biomolecules of interest, we first tested a series of standards (Table 1). The purchased or prepared standard solutions included: (1) a physiological amino acid standard (Sigma-Aldrich) containing 27 amino acids and related compounds in 0.1 M HCl; (2) a nucleobase standard solution containing seven nucleobases dissolved in 0.1 M NaOH; (3) two fatty acid standard solutions containing C12–C18 and C19–C26 fatty acids, respectively, dissolved in DCM; and (4) other standalone standard solutions, with glycolic acid, cysteine, asparagine, lysine, glutamine, ornithine, and tryptophan dissolved in 0.1 M HCl, while histidine and guanidine were dissolved in methanol. All solutions were prepared at 2.5 μ mol mL⁻¹ for each biomolecule at room temperature. 50 μ L of each standard solution was derivatized according to the protocol described above, and 1 μ L of the final solution was injected into a Thermo Scientific TSQ 9610 GC/MS/MS system equipped with a TG-5SILMS capillary column. Helium (99.9999%) was used as the carrier gas, with a constant flow rate of 1.3 mL/min. The oven temperature program began at 100 °C, ramped up at a rate of 10 °C/min, and reached a final temperature of 270 °C, which was held for 15 minutes. The temperatures of the GC inlet, transfer line, and ion source were set to 260, 280, and 300 °C, respectively.

The method development procedure is illustrated in Figure 1, using nucleobase standards as an example. Initially, the standard solution was analyzed in full-scan mode (Figure 1 (a)) to determine the retention time of each compound. The mass spectrum of each peak was compared with the standard spectra from the NIST database, and two parent ions

with both high intensity and large m/z values were selected ($m/z=307$ and 193), as shown in Figure 1 (b) for hypoxanthine. These parent ions were then fragmented under three different collision energies (10 eV, 20 eV, and 30 eV), and two representative daughter ions ($m/z=166$ and 84) with high intensity were selected for each parent ion, as shown in Figure 1 (c) for $m/z=193$. Subsequently, we optimized the collision energy for each characteristic ion pair to achieve maximum signal intensity (Figure 1 (d)). Method development and optimization were conducted using the AutoSRM software. The same procedures were applied to all 56 compounds listed in Table 1 for the method development (see Table A1 for the list of retention times and SRM transitions). Compared to previous studies (Sebree et al. 2018; Pearce et al. 2024a), we increased the number of characteristic ion pairs in our method to ensure more accurate and robust compound confirmation, while maintaining a low level of background noise.

Table 1. Standards Used for GC/MS/MS Method Development

Amino Acids	Alanine, Glycine, Sarcosine, Valine, Leucine, Isoleucine, Proline, Serine, Threonine, Phenylalanine, Methionine, Cysteine, Aspartic Acid, Hydroxyproline, Glutamic Acid, Asparagine, Lysine, Glutamine, Aminoadipic acid, Ornithine, Histidine, Tyrosine, Tryptophan, Cystine
Nucleobases	Uracil, Thymine, Cytosine, Hypoxanthine, Adenine, Xanthine, Guanine
Fatty Acids	Dodecanoic Acid (C12:0 ^a), Tridecanoic Acid (C13:0), Myristic Acid (C14:0), Pentadecanoic Acid (C15:0), Palmitic Acid (C16:0), Heptadecanoic Acid (C17:0), Stearic Acid (C18:0), Nondecanoic Acid (C19:0), Arachidic Acid (C20:0), Heneicosanoic Acid (C21:0), Behenic Acid (C22:0), Tricosanoic Acid (C23:0), Tetracosanoic Acid (C24:0), Pentacosanoic Acid (C25:0), Hexacosanoic Acid (C26:0), Palmitoleic Acid (C16:1 n-7), Linoleic Acid (C18:2 n-6), Oleic Acid (C18:1 n-9), α -Linolenic Acid (C18:3 n-3), γ -Linolenic Acid (C18:3 n-6), Isostearic Acid
Other Biomolecules	Glycolic Acid, Urea, Guanidine, Cystathionine

Note.

^a Conventional lipid nomenclature (C:D n-x), where C represents the total number of carbon atoms, D the number of double bonds, and n-x denotes the position of the first double bond counted from the terminal methyl end of the molecule.